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ass_17

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Python

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JupyterLab Python 3 (ipykernel)

```
[1]: import numpy as np
ram = np.array([4, 6, 8, 12, 16, 20, 24, 32]).reshape(-1, 1)
price = np.array([25000, 30000, 36000, 45000, 55000, 65000, 72000, 90000])
print("RAM (GB):")
print(ram)
print("\nLaptop Price (₹):")
print(price)

RAM (GB):
[[ 4]
 [ 6]
 [ 8]
[12]
[16]
[20]
[24]
[32]]

Laptop Price (₹):
[25000 30000 36000 45000 55000 65000 72000 90000]

[3]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    ram, price, test_size=0.25, random_state=42
)
print("Training Size:", len(X_train))
print("Testing Size:", len(X_test))

Training Size: 6
Testing Size: 2
```

Home x ass_17 x +

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JupyterLab Python 3 (ipykernel)

```
[4]: import pandas as pd
df = pd.DataFrame({
    "RAM(GB)": ram.flatten(),
    "Price(₹)": price
})
print(df)
print("\nDataset Shape:", df.shape)
```

	RAM(GB)	Price(₹)
0	4	25000
1	6	30000
2	8	36000
3	12	45000
4	16	55000
5	20	65000
6	24	72000
7	32	90000

Dataset Shape: (8, 2)

```
[5]: from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
```

Slope: 2294.117647058824
Intercept: 17127.450980392154

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JupyterLab Python 3 (ipykernel)

3	12	45000
4	16	55000
5	20	65000
6	24	72000
7	32	90000

Dataset Shape: (8, 2)

```
[5]: from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)

Slope: 2294.117647058824
Intercept: 17127.450980392154

[6]: pred_train = model.predict(X_train)
pred_test = model.predict(X_test)
new_ram = np.array([[18]])
predicted_price = model.predict(new_ram)
print("Predicted Laptop Price:", predicted_price[0])

Predicted Laptop Price: 58421.568627450986
```